

## Chapter 1: Intro to Tableau

- Use the filter shelf to add and customize filters for both measures and dimensions, such as "Cell Phones per 100 people" and "Broadband Subscribers per 100 people," to refine what data appears in your visualizations.
- Apply interactive filter cards to allow dynamic user interactions with your visualizations, enabling the selection of single or multiple values from a checklist format.
- Filter out null values to clean your data and avoid misleading visualizations that might incorrectly suggest zero values for certain data points.
- Implement TOP filters to focus on specific subsets of your data, like displaying only the top 5 countries based on the average "Cell Phones per 100 people" metric over a set period.

For example, to filter for non-null values in Tableau, you navigated to the special tab and selected non-null values, ensuring your visualizations only included meaningful data.

### Calculated fields

- **Creating Calculated Fields:** You discovered how to generate new data columns by writing formulas to perform operations on existing data. For instance, to analyze gender inequality in education across G20 countries, you subtracted average years in school for women from those for men, directly in Tableau without needing special functions.
- **Using Built-in Functions:** The lesson introduced Tableau's built-in functions like `POWER`, which raises a field to a specified power, enhancing your ability to manipulate data directly within your visualizations.
- **Rounding Data:** You learned to round data to create cleaner, more readable visualizations. For example, rounding the years of education data to the nearest decimal point improves the clarity of data labels.
- **Calculating Ratios:** Calculating ratios between genders for years spent in school was another exercise, showcasing how to compare two values effectively. A ratio close to one indicates equal levels of education between genders.
- **Averaging Across Genders:** Finally, you created a calculated field to find the average years of education across genders for the 25-34 age group, simplifying data comparison across countries.

Here's a snippet of how you might create a calculated field for averaging across genders:

```
([Average Years Men] + [Average Years Women]) / 2
```

### Reference Lines, Trend Lines, Forecasting

- **Reference Lines:** You discovered how to add a reference line to a line chart, using the average number of monthly births in Portugal as an example. This technique helps in

comparing individual data points against a common measure, making it easier to spot above or below average values.

- **Trend Lines:** You explored adding a linear trend line to visualize the overall direction (upward or downward) of births over time. This is particularly useful for identifying long-term trends in your data at a glance.
- **Forecasting:** You learned how to apply Tableau's forecasting feature to predict future values, such as the number of births in the next year. This predictive tool is invaluable for planning and decision-making based on historical data trends.

Here's a snippet of how you might add a trend line in Tableau:

Navigate to the Analytics pane -> Drag 'Trend Line' to the view -> Select 'linear'.

- **Creating an Area Chart:** You visualized Nintendo video game sales by release year to identify trends over time.
- **Sales by Platform:** You examined sales data across different platforms like Gameboy and Gameboy Advance.
- **Treemap Visualization:** You used a treemap to display sales information by genre, providing a clear hierarchical view.
- **Ranking Top Video Games:** You created a chart to rank the top-selling video games, highlighting the most popular titles.

You then combined these visualizations into a single dashboard:

- **Building the Dashboard:** You dragged and arranged the visualizations on a canvas, added legends and filters, and adjusted their positions.
- **Interactive Elements:** You enabled filters to make the dashboard interactive, allowing users to focus on specific data points.

Finally, you created a story by combining dashboards for Nintendo, Playstation, and Xbox, providing a comprehensive competitor analysis.

*# Example of adding a filter in a dashboard*

```
dashboard.add_filter('Genre', 'Simulation')
```

## Chapter 2: Analyzing Data in Tableau